Anvi Karpoor

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EDUCATION

University of California, San Diego - Cognitive Science with Specialization in Machine Learning & Neural Computation and Minor in Data Science | *Expected Graduation Date: Spring 2029*

Relevant Coursework: Principles of Data Science (DSC 10), Cognitive Consequences of Technology (COGS 10)

North Creek High School | *2021-2025*

Relevant Coursework: Intermediate Data Programming (CSE 163), Advanced Programming Topics, Advanced Robotics

EXPERIENCE/LEADERSHIP

Team Captain | FTC Robotics Team Omega Squad/RoboGOATS ***September 2019 - June 2024***

· Mentored 6 students in Java, teaching fundamentals, object-oriented programming, and command-based design

· Led design of drivetrains and mechanical subsystems, running failure mode analysis, and iteratively stress-testing

· Developed AR pipelines using Vuforia and TensorFlow, integrating them into autonomous routines for target tracking

· Utilized CAD tools (TinkerCAD) and fabrication workflows (rapid prototyping, iterative assembly, tolerance testing)

· Managed 3 community outreach programs, coordinating technical demonstrations, robot showcases, and STEM workshops while overseeing logistics and team operations

PROJECTS

Harvest Hollow | Java Farming Simulation Game ***March 2025 - June 2025***

· Built a custom Java game with a tile-based world, resource management, and procedural terrain generation using seeded randomization algorithms

· Engineered interactive systems for farming and inventory management; implemented state management and event-driven game logic

· Developed the full GUI layer with Java Swing, including sprite handling, collision detection, and animation timers

Computational Python Data Analysis | Gaming and Mental Health Report ***March 2024 - June 2024***

· Processed and validated data using Pandas, filtering functions, and automated tests with Python's unittest framework

· Developed visualizations with Vega-Altair, including regression plots and geospatial heatmaps using GeoPandas

· Applied statistical validation with SciPy to evaluate correlation strength and regression accuracy

Iris: Diabetes Monitoring Contact Lenses | TSA Engineering Design Project ***September 2023 - July 2024***

· Designed a non-invasive glucose monitoring contact lens using microsensors, microfluidics, and BLE communication

· Developed iterative prototypes using CAD modeling, optimizing sensor placement, ergonomics, and device reliability

· Mapped user experience flows for patients, including real-time alerts, provider data sharing, and customizable tracking

· Defined safety testing protocols for the sensor chemistry, tear sampling mechanism, and long-duration wear

Terra: Sensor System for Nitrogen Management | TSA Engineering Design Project ***September 2024 - April 2025***

· Designed a multi-depth soil sensing system using ion-selective, moisture, and temperature sensors

· Developed CAD models for the PVC housing and modular system architecture to support scalable field installation

· Created UI/UX requirements for farmer-facing dashboards focused on actionable nutrient and moisture data

HONORS & AWARDS

· **Engineering Design** - 5th Place, TSA Nationals

· **Future Technology and Engineering Teacher** - 9th Place, TSA Nationals

SKILLS

· **Languages & Tools:** Java (Swing) · Python (Pandas, SciPy) · TensorFlow · Vuforia Engine · FTC SDK · GDScript · TinkerCAD

· **Design & Visualization:** Figma · Adobe Suite · Canva · DaVinci Resolve · Godot · Microsoft Suite

· **Engineering & Quality:** Systems Design · Process Optimization · Technical Documentation · Data Analysis